

CUSTOMER SERVICE REQUEST ANALYSIS

Importing required libraries

Double-click (or enter) to edit

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Importing the dataset

```
df = pd.read_csv("/311_Service_Requests_from_2010_to_Present.csv")

/tmp/ipykernel_3442/1011216806.py:1: DtypeWarning: Columns (48,49) have mixed types. Specify dtype option on import or set low_memory=False.
df = pd.read_csv("/311_Service_Requests_from_2010_to_Present.csv")
```

Exploring the Data

```
from google.colab import drive
drive.mount('/content/drive')

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
df.head()
```

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor	Location Type	Incident Zip	Incident Address	...	Bridge Highway Name	Bridge Highway Direction	Road Ramp	Bridge Highway Segment	Garage Lot Name	Ferry Direction	Ferry Terminal Name	Latitude
0	32310363	12/31/2015 23:59	1/1/2016 0:55	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10034.0	71 VERMILYEA AVENUE	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	40.865682
1	32309934	12/31/2015 23:59	1/1/2016 1:26	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/Sidewalk	11105.0	27-07 23 AVENUE	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	40.775945
2	32309159	12/31/2015 23:59	1/1/2016 4:51	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/Sidewalk	10458.0	2897 VALENTINE AVENUE	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	40.870325
3	32305098	12/31/2015 23:57	1/1/2016 7:43	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking	Street/Sidewalk	10461.0	2940 BAISLEY AVENUE	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	40.835994
4	32306529	12/31/2015 23:56	1/1/2016 3:24	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk	Street/Sidewalk	11373.0	87-14 57 ROAD	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	40.733060
5 rows × 53 columns																			

```
df.columns
```

```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
      'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
      'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
      'Intersection Street 1', 'Intersection Street 2', 'Address Type',
      'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
      'Resolution Description', 'Resolution Action Updated Date',
      'Community Board', 'Borough', 'X Coordinate (State Plane)',
      'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
      'School Name', 'School Number', 'School Region', 'School Code',
      'School Phone Number', 'School Address', 'School City', 'School State',
      'School Zip', 'School Not Found', 'School or Citywide Complaint',
      'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
      'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
      'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
      'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location'],
      dtype='object')
```

```
df.shape
```

(364558, 53)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 364558 entries, 0 to 364557
Data columns (total 53 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Unique Key                           364558 non-null  int64
1   Created Date                          364558 non-null  object
2   Closed Date                           362177 non-null  object
3   Agency                                364558 non-null  object
4   Agency Name                           364558 non-null  object
5   Complaint Type                         364558 non-null  object
6   Descriptor                             358057 non-null  object
7   Location Type                          364425 non-null  object
8   Incident Zip                           361560 non-null  float64
9   Incident Address                       312859 non-null  object
10  Street Name                           312859 non-null  object
11  Cross Street 1                         307370 non-null  object
12  Cross Street 2                         306753 non-null  object
13  Intersection Street 1                   51120 non-null  object
14  Intersection Street 2                   50512 non-null  object
15  Address Type                           361306 non-null  object
16  City                                   361561 non-null  object
17  Landmark                               375 non-null    object
18  Facility Type                           362169 non-null  object
19  Status                                 364558 non-null  object
20  Due Date                               364555 non-null  object
21  Resolution Description                  364558 non-null  object
22  Resolution Action Updated Date          362156 non-null  object
23  Community Board                         364558 non-null  object
24  Borough                                364558 non-null  object
25  X Coordinate (State Plane)              360528 non-null  float64
26  Y Coordinate (State Plane)              360528 non-null  float64
27  Park Facility Name                      364558 non-null  object
28  Park Borough                           364558 non-null  object
29  School Name                             364558 non-null  object
30  School Number                           364558 non-null  object
31  School Region                           364557 non-null  object
32  School Code                             364557 non-null  object
33  School Phone Number                     364558 non-null  object
34  School Address                          364558 non-null  object
35  School City                             364558 non-null  object
36  School State                           364558 non-null  object
37  School Zip                              364557 non-null  object
38  School Not Found                        364558 non-null  object
39  School or Citywide Complaint            0 non-null     float64
40  Vehicle Type                            0 non-null     float64
41  Taxi Company Borough                    0 non-null     float64
42  Taxi Pick Up Location                    0 non-null     float64
43  Bridge Highway Name                     297 non-null    object
44  Bridge Highway Direction                 297 non-null    object
45  Road Ramp                               262 non-null    object
46  Bridge Highway Segment                  262 non-null    object
47  Garage Lot Name                         0 non-null     float64
48  Ferry Direction                          1 non-null     object
```

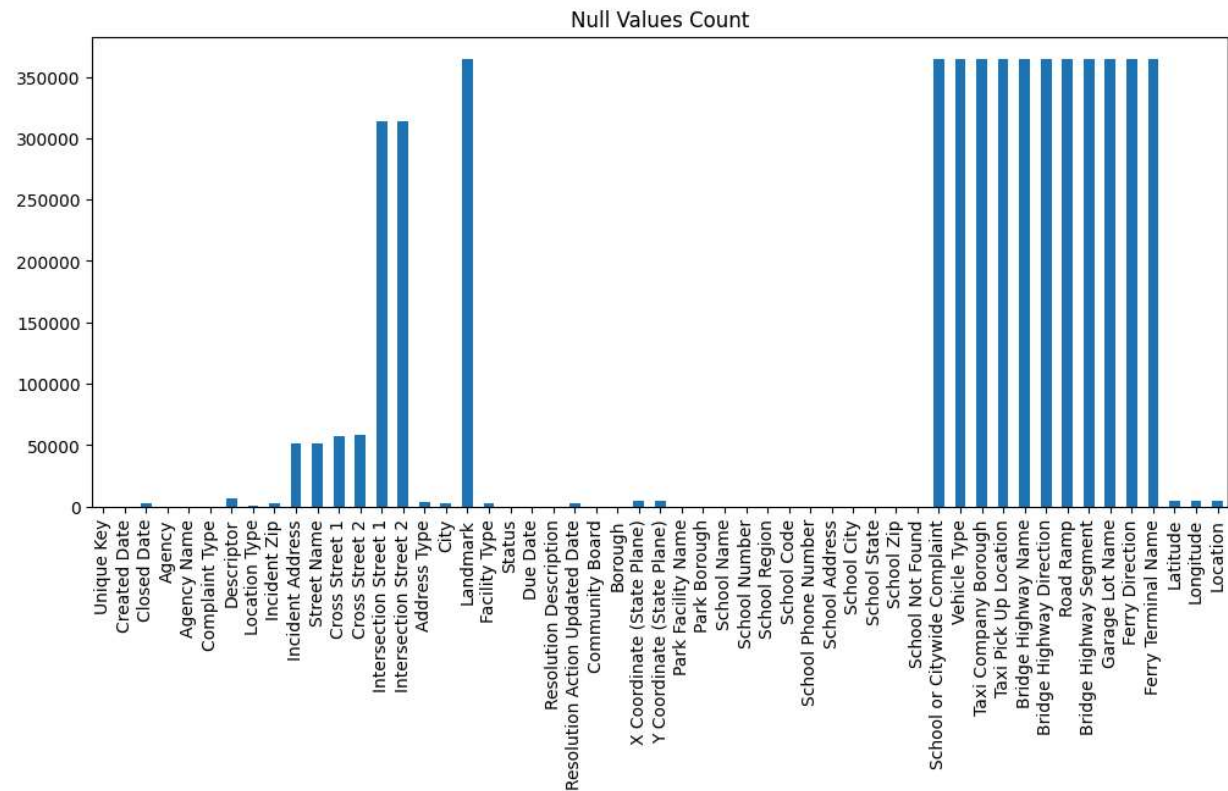
49	Ferry Terminal Name	2	non-null	object
50	Latitude	360528	non-null	float64
51	Longitude	360528	non-null	float64
52	Location	360528	non-null	object

df.isnull().sum()

	0
Unique Key	0
Created Date	0
Closed Date	2381
Agency	0
Agency Name	0
Complaint Type	0
Descriptor	6501
Location Type	133
Incident Zip	2998
Incident Address	51699
Street Name	51699
Cross Street 1	57188
Cross Street 2	57805
Intersection Street 1	313438
Intersection Street 2	314046
Address Type	3252
City	2997
Landmark	364183
Facility Type	2389
Status	0
Due Date	3
Resolution Description	0
Resolution Action Updated Date	2402
Community Board	0
Borough	0
X Coordinate (State Plane)	4030
Y Coordinate (State Plane)	4030
Park Facility Name	0
Park Borough	0
School Name	0
School Number	0
School Region	1
School Code	1
School Phone Number	0
School Address	0
School City	0
School State	0
School Zip	1
School Not Found	0
School or Citywide Complaint	364558
Vehicle Type	364558
Taxi Company Borough	364558
Taxi Pick Up Location	364558
Bridge Highway Name	364261
Bridge Highway Direction	364261
Road Ramp	364296
Bridge Highway Segment	364296
Garage Lot Name	364558
Ferry Direction	364557
Ferry Terminal Name	364556
Latitude	4030
Longitude	4030
Location	4030

dtype: int64

```
df.isnull().sum().plot(kind='bar', figsize=(12,5))
plt.title("Null Values Count")
plt.show()
```



```
df = df[df['Closed Date'].notnull()]
```

```
df['Created Date'] = pd.to_datetime(df['Created Date'])
df['Closed Date'] = pd.to_datetime(df['Closed Date'])
```

```
df['Request_Closing_Time'] = df['Closed Date'] - df['Created Date']
df['Request_Closing_Time'] = df['Request_Closing_Time'].dt.total_seconds()
```

```
df['Request_Closing_Time'].describe()
```

Request_Closing_Time	
count	3.621770e+05
mean	1.511326e+04
std	2.110254e+04
min	6.000000e+01
25%	4.560000e+03
50%	9.600000e+03
75%	1.890000e+04
max	2.134380e+06

dtype: float64

```
df['City'].fillna("Unknown City", inplace=True)
df['Complaint Type'].fillna("Unknown", inplace=True)
```

/tmp/ipykernel_3442/697721295.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

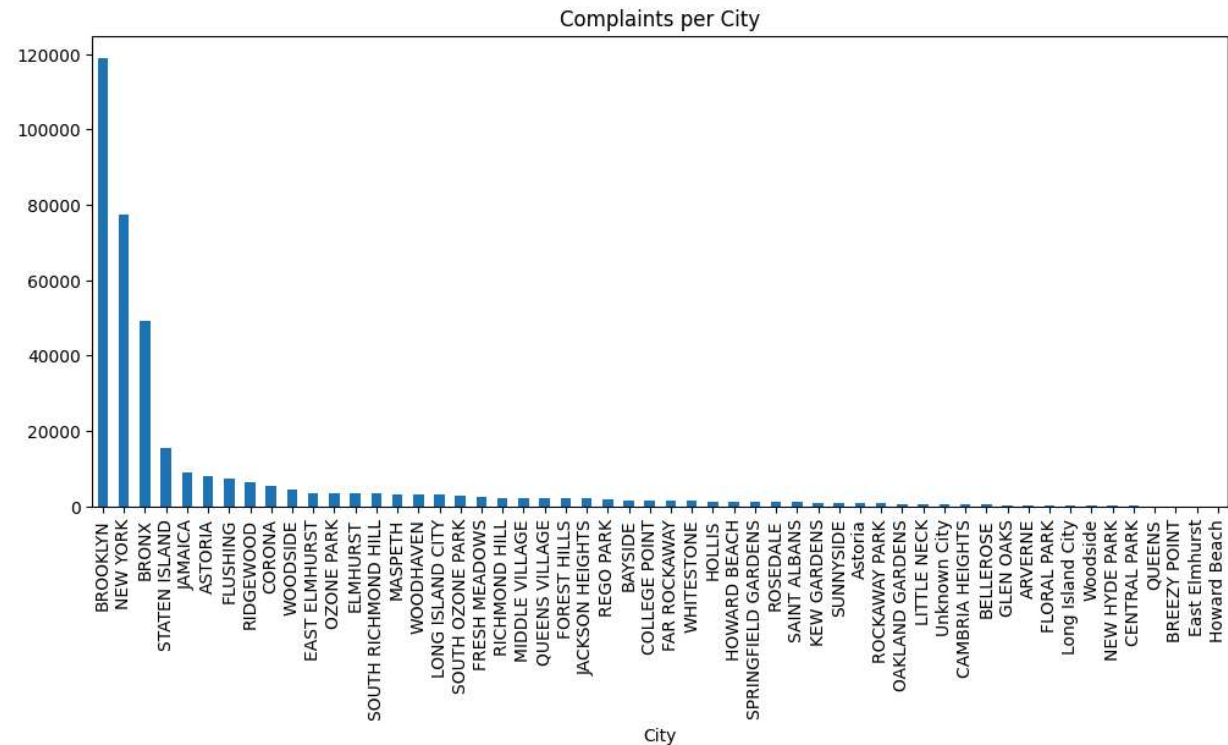
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on

```
df['City'].fillna("Unknown City", inplace=True)
/tmp/ipykernel_3442/697721295.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on

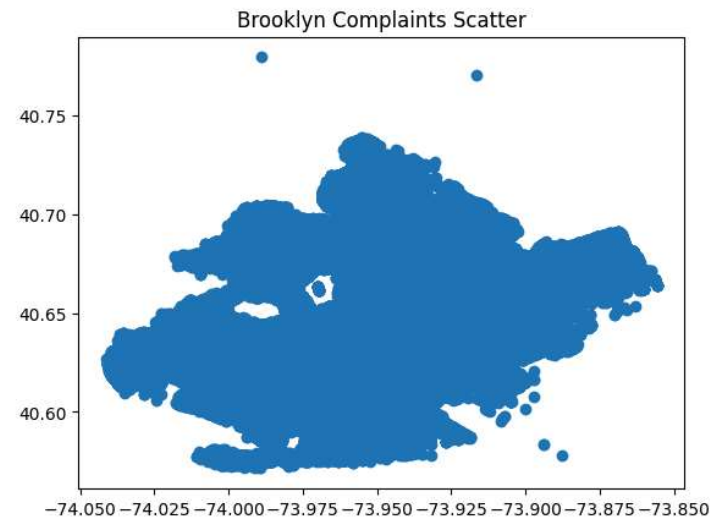
```
df['Complaint Type'].fillna("Unknown", inplace=True)
```

```
df['City'].value_counts().plot(kind='bar', figsize=(12,5))
plt.title("Complaints per City")
plt.show()
```

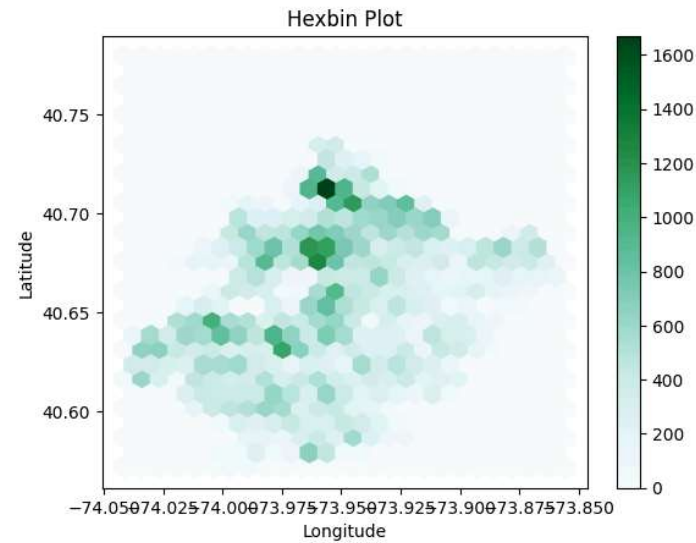


```
brooklyn = df[df['City'] == 'BROOKLYN']
```

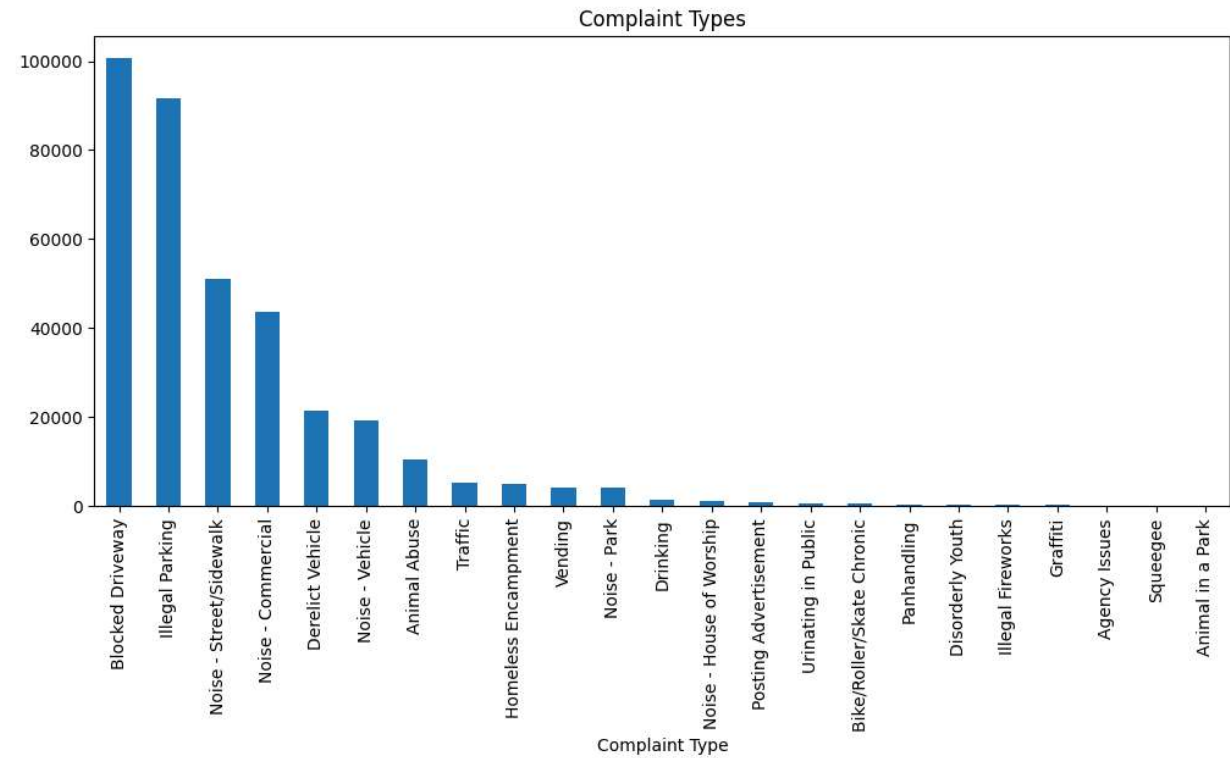
```
plt.scatter(brooklyn['Longitude'], brooklyn['Latitude'])
plt.title("Brooklyn Complaints Scatter")
plt.show()
```



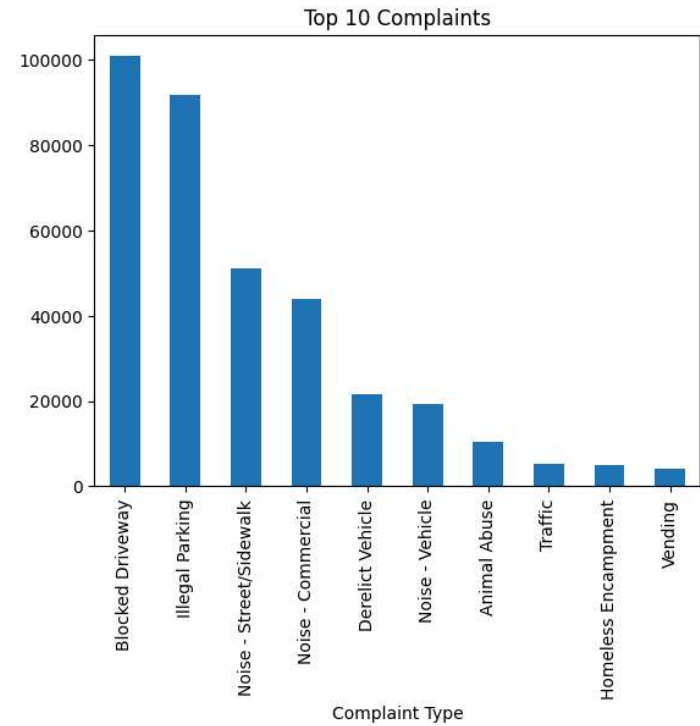
```
brooklyn.plot.hexbin(x='Longitude', y='Latitude', gridsize=25)
plt.title("Hexbin Plot")
plt.show()
```



```
df['Complaint Type'].value_counts().plot(kind='bar', figsize=(12,5))
plt.title("Complaint Types")
plt.show()
```



```
top10 = df['Complaint Type'].value_counts().head(10)
top10.plot(kind='bar')
plt.title("Top 10 Complaints")
plt.show()
```



```
df_new = pd.crosstab(df['Complaint Type'], df['City'])
df_new.head()
```

City	ARVERNE	ASTORIA	Astoria	BAYSIDE	BELLEROSE	BREEZY POINT	BRONX	BROOKLYN	CAMBRIA HEIGHTS	CENTRAL PARK	...	SOUTH OZONE PARK	SOUTH RICHMOND HILL	SPRINGFIELD GARDENS	STATEN ISLAND	SUNNYSIDE	Unknown City	WHITESTONE	WOODHAVEN	WOODSIDE
Complaint Type																				
Agency Issues	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	8	0	0	
Animal Abuse	46	170	0	53	15	2	1971	3191	15	0	...	74	40	42	786	40	1	43	57	
Animal in a Park	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	
Bike/Roller/Skate Chronic	0	16	0	0	1	0	22	124	0	0	...	1	1	0	10	2	2	4	2	
Blocked Driveway	50	3436	159	514	138	3	17062	36445	177	0	...	1202	1946	330	2845	278	86	279	1363	

5 rows × 54 columns

```
df_new.plot(kind='bar', stacked=True, figsize=(12,6))
plt.title("Complaints by City")
plt.show()
```

